Independent University, Bangladesh

CSE-313 Mid + Quiz Exam, Autumn-23, SET-A · Time: 1h30m

Real paper from this course. Quiz questions are on the other side of the paper.

Question paper

Mid Question (Total marks: 20)

1. Prove whether the following Grammar is Ambiguous or NOT. If the following grammar is ambiguous, convert it to unambiguous form. **[7]**

```
E → E @ S | S
S → S # B | B
B → C ~ B | C
C → C ? A | E
A → a
```

2. Write the First() and Follow() of the following Grammar. Draw LL(1) Parsing Table and Parse the string `( ) g c c d c e` showing all steps. If parsing is not possible, point out the reasons. Here, V = {E, U, V, W, X, Y}, T = {c, d, e, (,), g, ε}, and S = {E} **[8]**

```
E → c U V W | X Y E
U → c | ε
V → d U | ε
W → e | ε
X → e | ( Y )
Y → g | ε
```

3. Find True/ False of the following statements: **[5]**

(a) All TDP can use Left Recursive Grammar (LRG).
(b) All TDP can use Non-deterministic Grammar (NDG).
(c) From LRG grammar complete First set can't be found.
(d) From LRG grammar complete Follow set can't be found.
(e) Brute Force Parser is a BUP.

Quiz Question (Total marks: 10)

1. Convert the following Non Deterministic Grammar (NDG) to Deterministic Grammar (DG): **[5]**

```
S → b a c d S | b a c P | b a Q | b a
P → c P | c
Q → a Q | a
```

2. Convert the following Left Recursive Grammar (LRG) to Right Recursive Grammar (RRG): **[5]**

```
A → A % B | B
B → B & C | C
C → C ^ 0 1 | a b c
```